University of Applied Sciences

HOCHSCHULE EMDEN·LEER

Top 10 Machine Learning Algorithms

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Agenda

1. Naïve Bayes Classifier Algorithm
2. K-Means Clustering Algorithm
3. Support Vector Machine Learning Algorithm
4. Apriori Machine Learning Algorithm
5. Linear Regression Machine Learning Algorithm
6. Logistic Regression Machine Learning Algorithm
7. Decision Tree Machine Learning Algorithm
8. Random Forest Machine Learning Algorithm
9. Artificial Neural Networks Machine Learning Algorithm
10. K-Nearest Neighbors Algorithm

Top 10 - Machine Learning Algorithms

Source: www.dezyre.com

Top 10 - Machine Learning Algorithms

Machine Learning Algorithms Every Engineer must know:

1. Naïve Bayes Classifier Algorithm
2. K-Means Clustering Algorithm
3. Support Vector Machine Algorithm
4. Apriori Algorithm
5. Linear Regression
6. Logistic Regression
7. Decision Tree
8. Random Forest
9. Artificial Neural Networks
10. Nearest Neighbours

Machine Learning algorithms' classification

Machine Learning algorithms are classified as:

1. **Supervised Machine Learning Algorithms**
   Machine learning algorithms that make predictions on given set of samples. Supervised machine learning algorithm searches for patterns within the value labels assigned to data points.
2. **Unsupervised Machine Learning Algorithms**
   There are no labels associated with data points. These machine learning algorithms organize the data into a group of clusters to describe its structure and make complex data look simple and organized for analysis.
3. **Reinforcement Machine Learning Algorithms**
   These algorithms choose an action, based on each data point and later learn how good the decision was. Over time, the algorithm changes its strategy to learn better and achieve the best reward.

Naïve Bayes Classifier Algorithm

Naïve Bayes Classifier Algorithm

It would be difficult and practically impossible to classify a web page, a document, an email or any other lengthy text notes manually. This is where Naïve Bayes Classifier machine learning algorithm comes to the rescue.

A **classifier** is a function that allocates a population's element value from one of the available categories.

For instance, Spam Filtering is a popular application of Naïve Bayes algorithm. Spam filter here, is a classifier that assigns a label "Spam" or "Not Spam" to all the emails.

Naïve Bayes Classifier Algorithm

Naïve Bayes Classifier is amongst the most popular learning method grouped by **similarities**, that works on the popular Bayes Theorem of Probability - to build machine learning models particularly for disease prediction and document classification.

It is a **simple classification** of words based on **Bayes Probability Theorem** for subjective analysis of content.

When to use Naïve Bayes Classifier algorithm?

- If you have a moderate or large training data set.
- If the instances have several attributes.
- Given the classification parameter, attributes which describe the instances should be conditionally independent.

Applications of Naïve Bayes Classifier

Source: www.dezyre.com

Applications of Naïve Bayes Classifier

1. **Sentiment Analysis**: It is used at Facebook to analyse status updates expressing positive or negative emotions.
2. **Document Categorization**: Google uses document classification to index documents and find relevancy scores i.e. the PageRank. PageRank mechanism considers the pages marked as important in the databases that were parsed and classified using a document classification technique.
3. Naïve Bayes Algorithm is also used for **classifying new articles** about Technology, Entertainment, Sports, Politics, etc.
4. **Email Spam Filtering**: Google Mail uses Naïve Bayes algorithm to classify your emails as Spam or Not Spam

Advantages of the Naïve Bayes Classifier Algorithm

1. Naïve Bayes Classifier algorithm performs well when the input variables are **categorical**
2. A Naïve Bayes classifier **converges faster**, requiring relatively little training data than other discriminative models like logistic regression, when the **Naïve Bayes conditional independence assumption** holds.
3. With Naïve Bayes Classifier algorithm, it is easier to predict class of the test data set. A good bet for **multi class predictions** as well.
4. Though it requires conditional independence assumption, Naïve Bayes Classifier has presented good performance in **various application domains**.

K-Means Clustering Algorithm

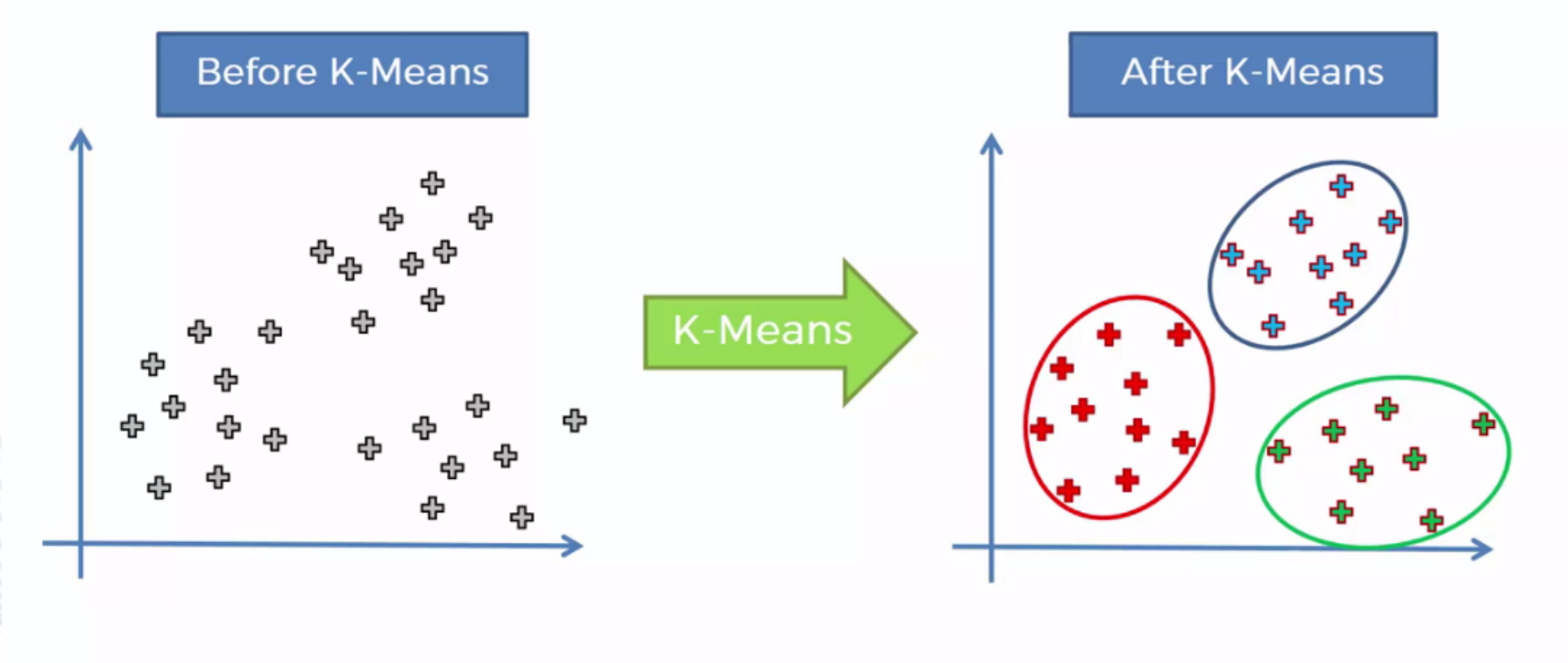
K-Means Clustering Algorithm

K-Means is a popularly used **unsupervised machine** learning algorithm for **cluster analysis**. K-Means is a non-deterministic and iterative method. The algorithm operates on a given data set through **pre-defined number of clusters, k**. The output of K-Means algorithm is k clusters with input data partitioned among the clusters.

K-Means Clustering Algorithm - Example

Let's consider K-Means Clustering for Wikipedia Search results. The search term "Jaguar" on Wikipedia will return all pages containing the word Jaguar which can refer to Jaguar as a Car, Jaguar as Mac OS version and Jaguar as an Animal. K-Means clustering algorithm can be applied to group the webpages that talk about similar concepts. So, the algorithm will group all web pages that talk about Jaguar as an Animal into one cluster, Jaguar as a Car into another cluster and so on.

Application of K-Means Clustering Algorithm



Source: towardsdatascience.com

Advantages of using K-Means Clustering ML Algorithm

1. In case of **globular clusters**, K-Means produces tighter clusters than **hierarchical clustering**.
2. Given a **smaller value of K**, K-Means clustering computes **faster** than hierarchical clustering for large number of variables.

Applications of K-Means Clustering

K-Means Clustering algorithm is used by **most of the search engines** like Yahoo, Google to cluster web pages by similarity and identify the "relevance rate" of search results. This helps search engines reduce the computational time for the users.

Support Vector Machine Learning Algorithm

Support Vector Machine Learning Algorithm

Support Vector Machine is a **supervised machine learning** algorithm for **classification** or **regression** problems where the dataset teaches SVM about the classes so that SVM can classify any new data.

It works by classifying the data into different classes by finding a line (**hyperplane**) which separates the training data set into classes. As there are many such linear hyperplanes, SVM algorithm tries to maximize the distance between the various classes that are involved and this is referred as margin maximization. If the line that maximizes the distance between the classes is identified, the probability to generalize well to unseen data is increased.

SVM's are classified into two categories

1. **Linear SVM's**: In linear SVM's the training data i.e. classifiers are separated by a hyperplane.
2. **Non-Linear SVM's**: In non-linear SVM's it is not possible to separate the training data using a hyperplane. For example, the training data for Face detection consists of group of images that are faces and another group of images that are not faces (in other words all other images in the world except faces). Under such conditions, the training data is too complex that it is impossible to find a representation for every feature vector. Separating the set of faces linearly from the set of non-face is a complex task.

Advantages of Using SVM

1. SVM offers best **classification performance** (accuracy) on the training data.
2. SVM renders more **efficiency** for correct classification of the future data.
3. The best thing about SVM is that it does **not** make any **strong assumptions** on data.
4. It does **not overfit** the data.

Applications of Support Vector Machine

SVM is commonly used for stock market forecasting by various financial institutions. For instance, it can be used to compare the relative performance of the stocks when compared to performance of other stocks in the same sector. The **relative comparison of stocks** helps manage investment making decisions based on the classifications made by the SVM learning algorithm.

Apriori Machine Learning Algorithm

Apriori Machine Learning Algorithm

Apriori algorithm is an **unsupervised machine learning algorithm** that generates **association rules** from a given data set.

Association rule implies that if an item A occurs, then item B also occurs with a certain probability. Most of the association rules generated are in the **IF_THEN** format. For example, **IF** people buy an iPad **THEN** they also buy an iPad Case to protect it. For the algorithm to derive such conclusions, it first observes the number of people who bought an iPad case while purchasing an iPad. This way a ratio is derived like out of the 100 people who purchased an iPad, 85 people also purchased an iPad case.

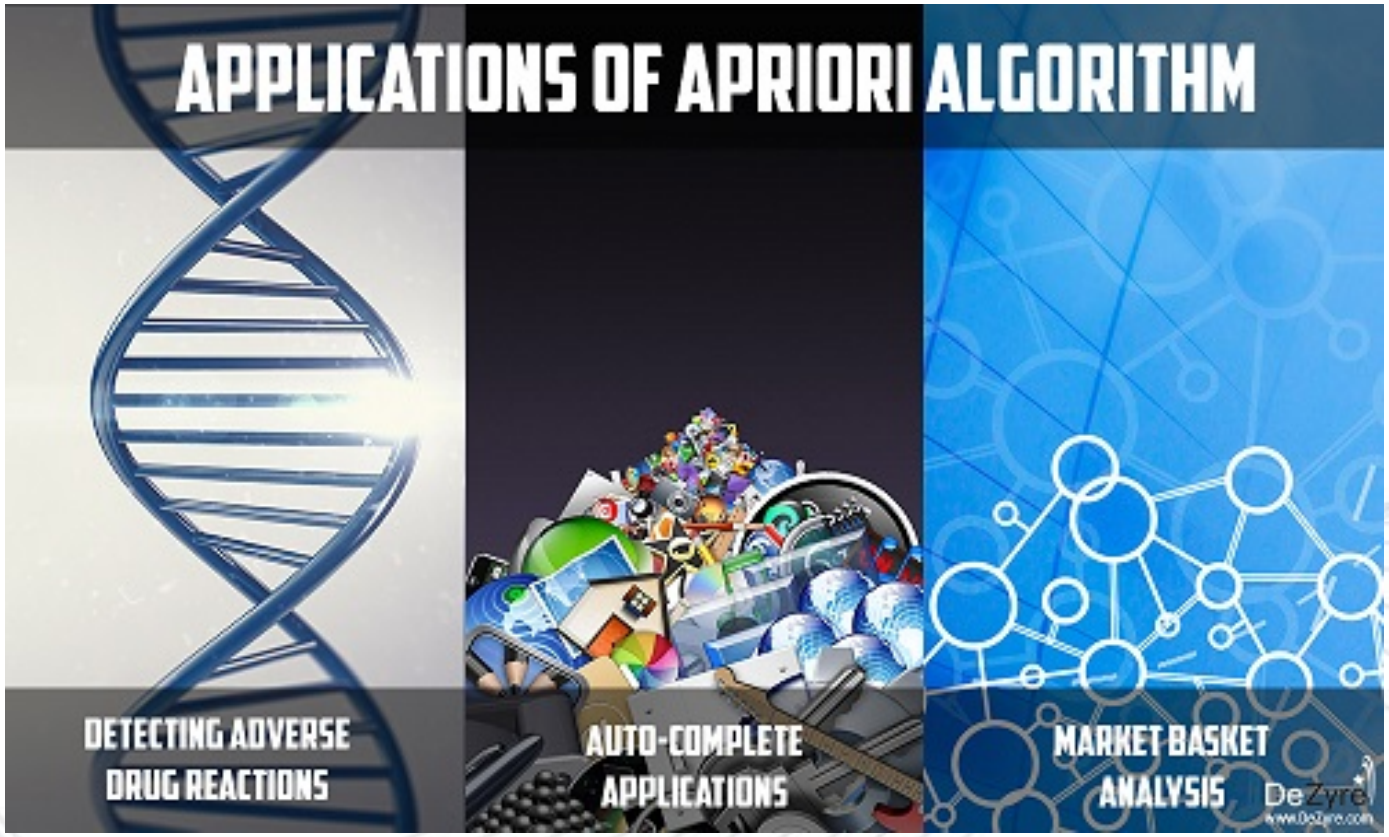
Principle on which Apriori ML Algorithm works:

1. If an item set occurs **frequently** then all the **subsets** of the item set, also occur **frequently**.
2. If an item set occurs **infrequently** then all the **supersets** of the item set have **infrequent** occurrence.

Advantages of Apriori Algorithm

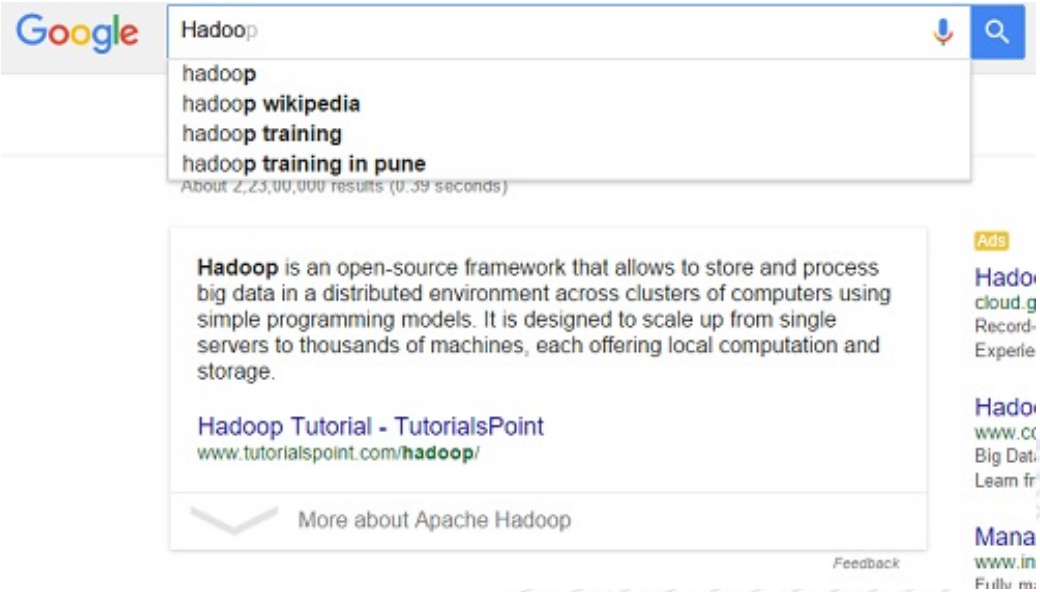
1. It is **easy to implement**, and
2. can **be parallelized** easily.
3. Apriori implementation makes use of **large item set properties**.

Applications of Apriori Algorithm



Source: www.dezyre.com

Applications of Apriori Algorithm



Source: www.dezyre.com

Applications of Apriori Algorithm

Detecting Adverse Drug Reactions Apriori algorithm is used for association analysis on healthcare data like-the drugs taken by patients, characteristics of each patient, adverse ill-effects patients experience, initial diagnosis, etc. This analysis produces association rules that help identify the combination of patient characteristics and medications that lead to adverse side effects of the drugs.

Market Basket Analysis Many e-commerce giants like Amazon use Apriori to draw data insights on which products are likely to be purchased together and which are most responsive to promotion. For example, a retailer might use Apriori to predict that people who buy sugar and flour are likely to buy eggs to bake a cake.

Applications of Apriori Algorithm

Auto-Complete Applications Google auto-complete is another popular application of Apriori wherein - when the user types a word, the search engine looks for other associated words that people usually type after a specific word.

Linear Regression Machine Learning Algorithm

Linear Regression Machine Learning Algorithm

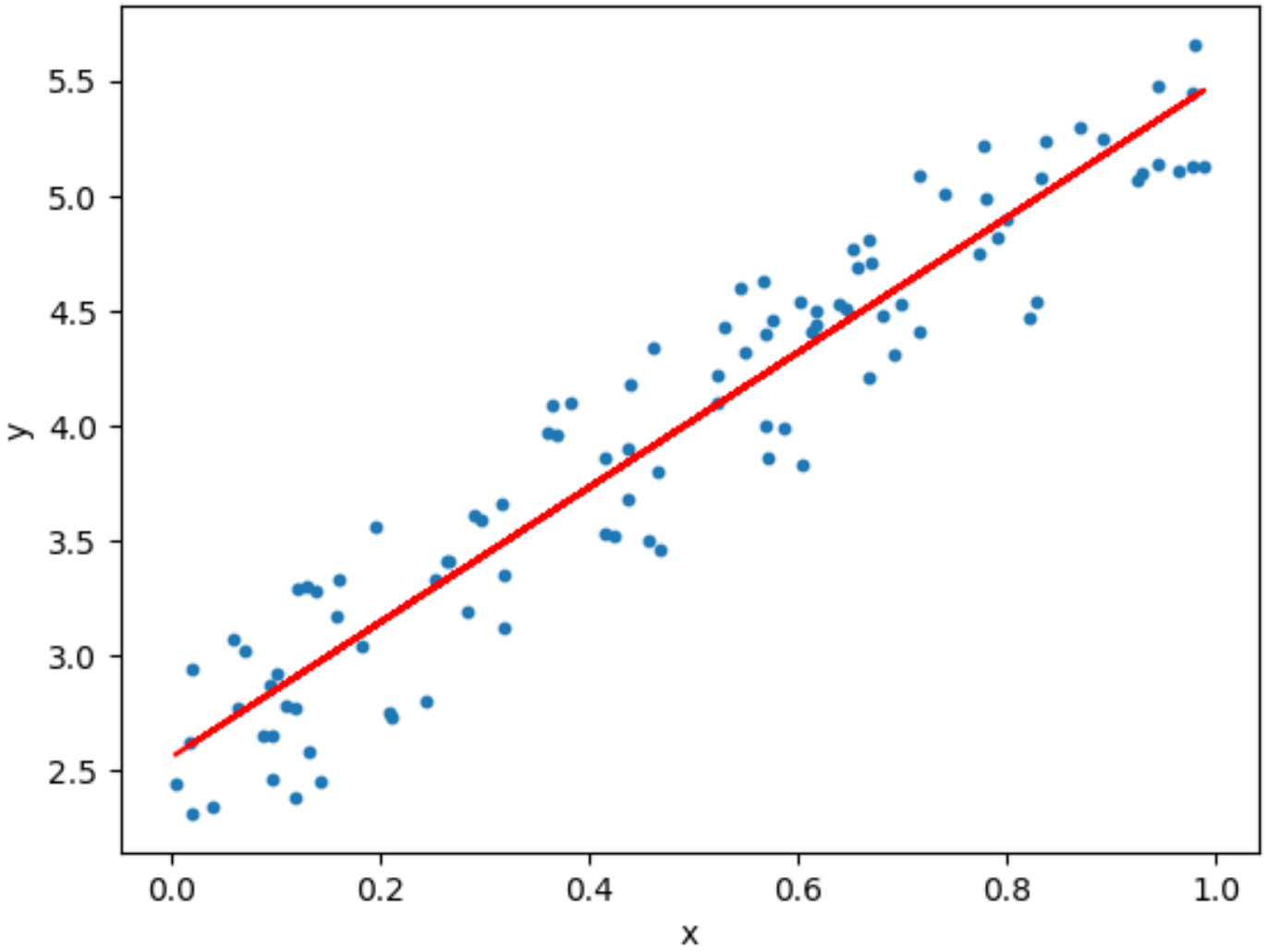
Linear Regression algorithm shows the **relationship between 2 variables** and how the change in one variable impacts the other.

The algorithm shows the impact on the dependent variable on changing the independent variable. The independent variables are referred as explanatory variables, as they explain the factors the impact the dependent variable. Dependent variable is often referred to as the factor of interest or predictor.

Advantages of Linear Regression ML Algorithm

1. It is one of the **most interpretable** machine learning algorithms, making it easy to explain to others.
2. It is **easy of use** as it requires minimal tuning.
3. It is the **mostly widely used** machine learning technique that runs fast.

Applications of Linear Regression ML Algorithm



Applications of Linear Regression ML Algorithm

Estimating Sales Linear Regression finds great use in business, for sales forecasting based on the trends. If a company observes steady increase in sales every month - a linear regression analysis of the monthly sales data helps the company forecast sales in upcoming months.

Risk Assessment Linear Regression helps assess risk involved in insurance or financial domain. A health insurance company can do a linear regression analysis on the number of claims per customer against age. This analysis helps insurance companies find, that older customers tend to make more insurance claims. Such analysis results play a vital role in important business decisions and are made to account for risk.

Logistic Regression Machine Learning Algorithm

Logistic Regression Machine Learning Algorithm

The name of this algorithm could be a little confusing in the sense that Logistic Regression machine learning algorithm is for **classification** tasks and not regression problems.

$\rightarrow$ The name "Regression" here implies that a **linear model** is fit into the feature space.

This algorithm applies a **logistic function** to a linear combination of features to predict the outcome of a categorical dependent variable based on predictor variables.

Logistic Regression Machine Learning Algorithm

The odds or probabilities that describe the outcome of a single trial are modelled as a function of explanatory variables. Logistic regression algorithms helps **estimate the probability** of falling into a specific level of the categorical dependent variable based on the given predictor variables.

Just suppose that you want to predict if there will be a snowfall tomorrow in New York. Here the outcome of the prediction is not a continuous number because there will either be snowfall or no snowfall and hence linear regression cannot be applied.

Types of Logistic Regression

- Binary Logistic Regression
- Multi-nominal Logistic Regression
- Ordinal Logistic Regression

Types of Logistic Regression - Binary Logistic Regression

- The most commonly used logistic regression when the categorical response has 2 possible outcomes i.e. either yes or not.
- Examples:
  - Predicting whether a student will pass or fail an exam.
  - Predicting whether a student will have low or high blood pressure.
  - Predicting whether a tumour is cancerous or not.

Types of LR - Multi-nominal Logistic Regression

- Categorical response has 3 or more possible outcomes with no ordering.
- Example
  - Predicting what kind of search engine (Yahoo, Bing, Google, and MSN) is used by majority of US citizens.

Types of Logistic Regression - Ordinal Logistic Regression

- Categorical response has 3 or more possible outcomes with natural ordering.
- Example:
  - How a customer rates the service and quality of food at a restaurant based on a scale of 1 to 10.

When to Use Logistic Regression ML Algorithm

1. Use logistic regression algorithms when there is a requirement to model the probabilities of the response variable as a function of some other explanatory variable.
2. Use logistic regression algorithms when there is a need to predict probabilities that categorical dependent variable will fall into two categories of the binary response as a function of some explanatory variables.
3. Logistic regression algorithms is best suited when the need is to classify elements two categories based on the explanatory variable.

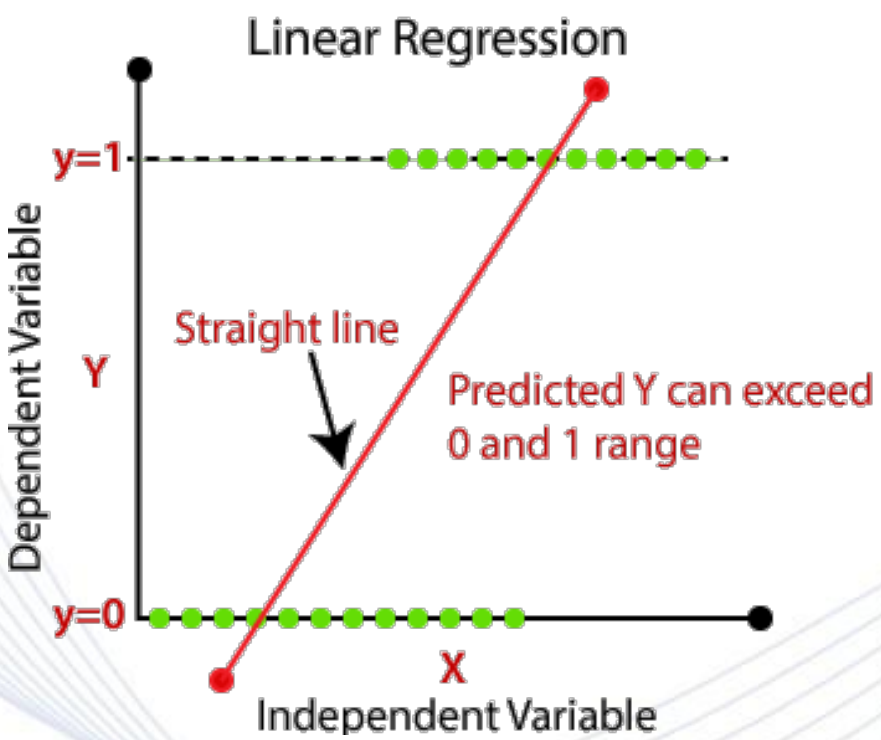
Advantages of Using Logistic Regression

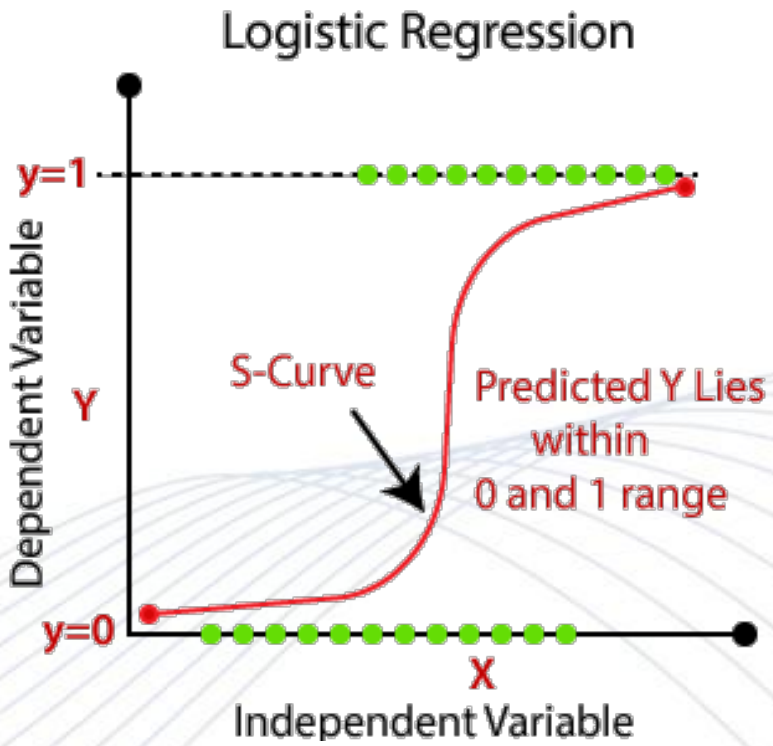
1. **Easier to inspect** and less complex.
2. **Robust algorithm** as the independent variables need not have equal variance or normal distribution.
3. These algorithms do **not assume a linear relationship** between the dependent and independent variables and hence can also handle non-linear effects.
4. Controls confounding and tests interaction.

Drawbacks of Using Logistic Regression

1. When the training data is **sparse** and **high dimensional**, in such situations a logistic model may over fit the training data.
2. They **cannot predict continuous** outcomes.
3. They **require more data** to achieve stability and meaningful results. These algorithms require minimum of 50 data points per predictor to achieve stable outcomes.
4. It predicts outcomes depending on a group of independent variables; if the **identification** of the independent variables is **wrong**, then the developed model will have minimal or no predictive value.
5. It is **not robust to outliers** and missing values.

Applications of Linear Regression ML Algorithm





Source: https://flipboard.com/@nehamail

Applications of Logistic Regression

1. Logistic regression algorithm is applied in the field of epidemiology to identify risk factors for diseases and plan accordingly for preventive measures.
2. Used to predict whether a candidate will win or lose a political election or to predict whether a voter will vote for a particular candidate.
3. Used to classify a set of words as nouns, pronouns, verbs, adjectives.
4. Used in weather forecasting to predict the probability of rain.
5. Used in credit scoring systems for risk management to predict the defaulting of an account.

Decision Tree Machine Learning Algorithm

Decision Tree Machine Learning Algorithm

A decision tree is a **graphical representation** that makes use of branching methodology to exemplify all possible outcomes of a decision, based on certain conditions.

In a decision tree, the internal node represents a test on the attribute, each branch of the tree represents the outcome of the test and the leaf node represents a **particular class label** i.e. the decision made after computing all of the attributes.

The classification rules are represented through the path from root to the leaf node.

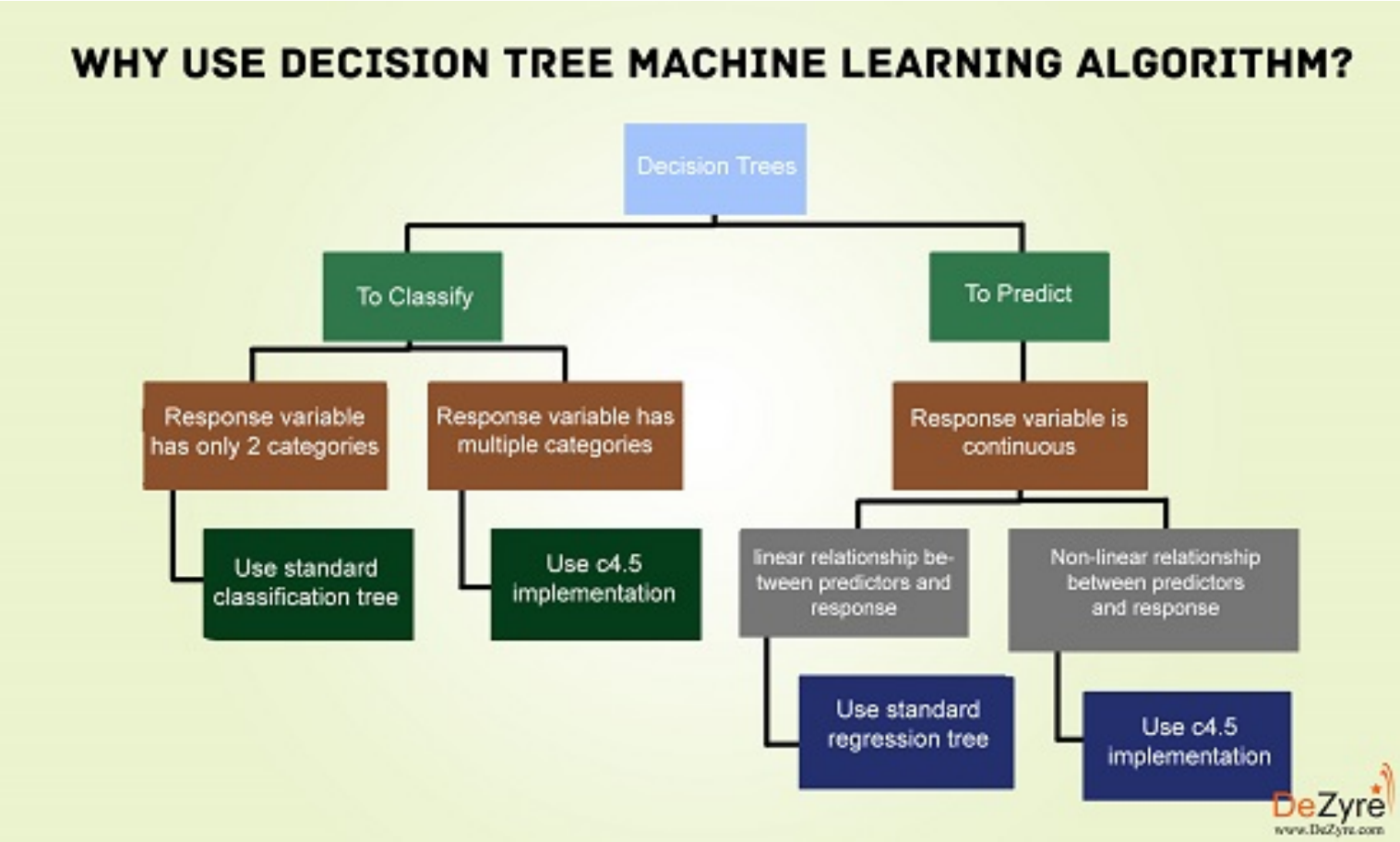
Types of Decision Trees

Classification Trees: These are considered as the default kind of decision trees used to separate a dataset into different classes, based on the response variable. These are generally used when the response variable is categorical in nature.

Regression Trees: When the response or target variable is continuous or numerical, regression trees are used. These are generally used in predictive type of problems when compared to classification.

It is the target variable that helps decide what kind of decision tree would be required for a particular problem.

Decision Tree Machine Learning Algorithm



Source: www.dezyre.com

Why should you use Decision Tree ML algorithm?

1. They help make decisions **under uncertainty** and help you **improve communication**, as they present a visual representation of a decision situation.
2. Decision tree machine learning algorithms help a data scientist capture the idea that if a **different decision** was taken, then how the operational nature of a situation or model would have **changed intensely**.
3. Decision tree algorithms help make **optimal decisions** by allowing to traverse through forward and backward calculation paths.

When to use Decision Tree Machine Learning Algorithm

1. Decision trees are **robust to errors** and if the training data contains errors - decision tree algorithms will be best suited to address such problems.
2. Decision trees are **best suited** for problems where instances are represented by **attribute value pairs**.
3. If the training data has **missing value** then decision trees can be used, as they **can handle** missing values nicely by looking at the data in other columns.
4. They are **best suited** when the target function has **discrete output values**.

Advantages of Using Decision Tree ML Algorithms

1. Decision trees are very **instinctual** and can be **explained** to anyone with ease.
2. They can **handle categorical** and **numerical** variables.
3. They do **not require** making any assumption on the **linearity** in the data.
4. They do **not** make any **assumptions** on the **classifier structure** and **space distribution**.
5. These algorithms are useful in data exploration: They **implicitly perform feature selection** which is very important in predictive analytics.
6. They are **not sensitive** to **missing values** and **outliers**.
   - Missing values will **not stop** you from splitting the data for building a decision tree.
   - Outliers will **not affect** the decision trees as data splitting happens based on some samples within the split range and not on exact absolute values.

Drawbacks of Using Decision Tree ML Algorithms

1. The **more the number** of decisions in a tree, **less** is the **accuracy** of any expected outcome.
2. Their outcomes may be **based on expectations**: When decisions are made in **real-time**, the payoffs and resulting outcomes might not be the same as expected or planned.
3. Any **irrational expectations** could lead to **major errors** in decision tree analysis.
4. It is **not always possible** to plan for **all eventualities** that can arise from a decision.
5. They do **not fit** well for **continuous** variables and result in instability and classification plateaus.

Drawbacks of Using Decision Tree ML Algorithms

- They are **easy to use** when compared to other decision making models.
- Creating **large decision trees** that contain several branches is a **complex** and **time consuming** task.
- They consider **only one attribute at a time** and might **not be best suited** for **actual data** in the decision space.
- **Large** sized decision trees with multiple branches are **not comprehensible** and pose several presentation difficulties.

Applications of Decision Tree ML Algorithm

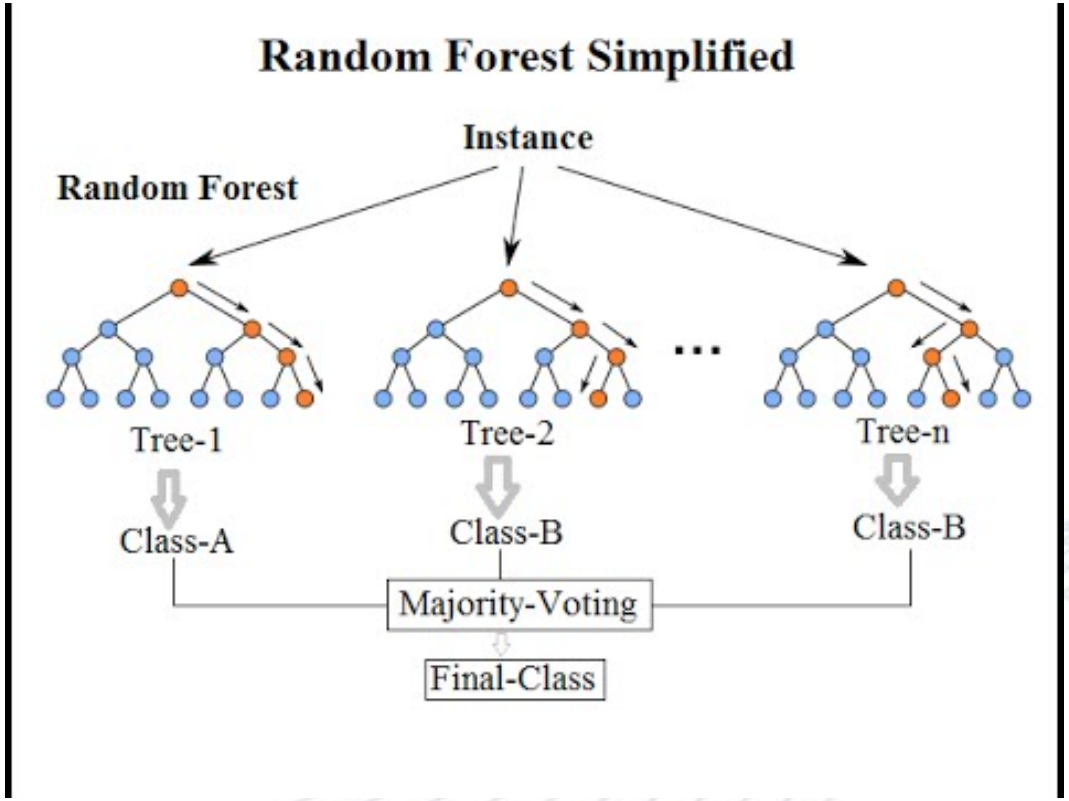
1. They find great use in finance for option pricing.
2. Remote sensing is an application area for pattern recognition based on decision trees.
3. They are used by banks to classify loan applicants by their probability of defaulting payments.
4. Gerber Products, a baby product company, used decision tree machine learning algorithm to decide whether they should continue using the plastic PVC (Poly Vinyl Chloride) in their products.
5. Rush University Medical Centre has developed a tool named Guardian that uses a decision tree machine learning algorithm to identify at-risk patients and disease trends.

Random Forest Machine Learning Algorithm

Random Forest Machine Learning Algorithm

- Random Forest is the go to machine learning algorithm that uses a **bagging approach** to create a bunch of decision trees with random subset of the data.
- A model is **trained several times** on random sample of the dataset to achieve good prediction performance from the random forest algorithm.
- In this ensemble learning method, the output of all the decision trees in the random forest, is **combined to make the final prediction**.
- The final prediction of the random forest algorithm is derived by **polling** the results of **each** decision tree or just by going with a prediction that appears the **most times** in the decision trees.

Random Forest Machine Learning Algorithm



Source: www.kdnuggets.com

Why use Random Forest Machine Learning Algorithm?

1. There are **many good open source**, free implementations of the algorithm available in Python and R.
2. They are **easy to implement**.
3. It maintains **accuracy** when there is **missing data** and is also **resistant to outliers**.
4. They are capable of handling **numerical**, **binary** and **categorical** features, without scaling, transformation or modification.
   $\Rightarrow$ Random Forest machine learning algorithms help data scientists save data preparation time.
5. **Implicit feature selection** as it gives estimates on what variables are **important in the classification**.

Advantages of Using Random Forest ML Algorithm

1. Random Forest is one of the **most effective** and **versatile** machine learning algorithm for wide variety of **classification** and **regression** tasks.
2. **Overfitting** is **less** of an issue with Random Forests.
3. These algorithms are **fast** but not in all cases.
4. They are more **robust to noise**.
5. One can **easily build** a decent model without much tuning.
6. Random Forest machine learning algorithms can be grown in **parallel**.
7. This algorithm runs **efficiently on large databases**.
8. Has higher **classification accuracy**.

Drawbacks of Using Random Forest ML Algorithm

1. Their **theoretical analysis** is **difficult**.
2. **Large** number of decision trees in the random forest can **slow down** the algorithm in making real-time predictions.
3. If the data consists of categorical variables with **different number of levels**, then the algorithm gets **biased** in favour of those attributes that have more levels.
4. Variable **importance scores** do **not** seem to be **reliable**.
5. When using RandomForest algorithm for regression tasks, it does **not predict beyond** the range of the response values in the **training data**.

Applications of Random Forest ML Algorithm

1. They are used by banks to predict if a loan applicant is a likely high risk.
2. They are used in the automobile industry to predict the failure or breakdown of a mechanical part.
3. They are used in the healthcare industry to predict if a patient is likely to develop a chronic disease or not.
4. They are used for regression tasks like predicting the average number of social media shares and performance scores.
5. They are used in predicting patterns, in speech recognition software and classifying images and texts.

Artificial Neural Networks
Machine Learning Algorithm

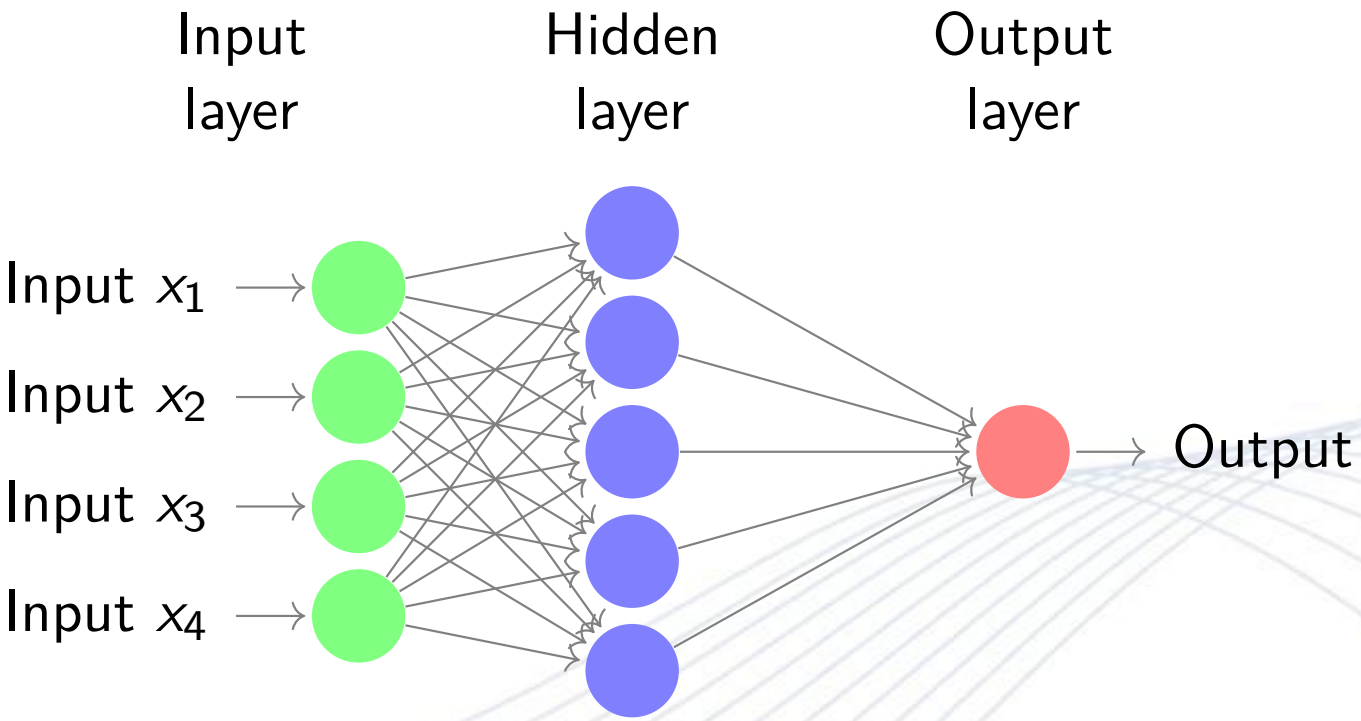
Artificial Neural Networks Machine Learning Algorithm

- Names:
  - Artificial Neural Network (ANN)
  - Deep Neural Network (DNN)
- It's a supervised machine learning method.

Artificial Neural Networks Machine Learning Algorithm

A neural network is an information technology system that is based on the human brain and equips computers with features of artificial intelligence.

ANN Machine Learning Algorithm - Structure



Applications

- Control and analysis of complex processes
- Error detection
- Optimization
- Time series analysis (weather, stocks, etc.)
- Image processing and pattern recognition
- Character recognition (OCR)

K-Nearest Neighbors Algorithm

Advantages

1. The k-nearest neighbors (KNN) algorithm is a **supervised machine learning algorithm**.
2. The algorithm can be used to solve both **classification** and **regression** problems.
3. The algorithm is **simple** and **easy to implement**.
4. There's **no need** to build a **model**, **tune** several parameters, or make **additional assumptions**.
5. The algorithm is **versatile**. It can be used for classification, regression, and search.

K-Nearest Neighbors Algorithm

KNN works:

1. Caluclating the distances between a query and all given data.
2. Selecting the specified number examples (K) closest to the query.
3. Result is
   - the most frequent label in the case of classification or
   - the average of the labels in the case of regression.

Disadvantages

The algorithm gets significantly slower as the number of examples and/or predictors/independent variables increase.